FORMULA 1 PODIUM PREDICTION COMPANION

Milestone 1 | Analytical report | ACL_Team70

Project summary

This project builds a leakage-aware driver-race dataset and evaluates whether a driver will finish in the top three using information available after qualifying and before the race. The analysis also answers three descriptive questions about circuit conversion, home-country starts, and constructor mechanical retirements.

Key findings

- **Data:** 8,970 driver-race rows from 2003-2024; 14.3% are podium finishes. The modeling key is unique.
- **Prediction:** The shallow neural network achieved the best validation PR-AUC (0.734); its held-out test PR-AUC was 0.679.
- **Most useful feature group:** removing grid and qualifying reduced the logistic baseline validation PR-AUC by 0.088.
- **Interpretation:** model explanations identify predictive signals; they do not establish causal effects.

Prediction target and information cutoff

The target is 1 when the driver finished with positionOrder <= 3 and 0 otherwise. Current-race finishing position, status, points, laps, fastest lap, pit stops, and post-race standings are excluded from model features because they are known only after the race.

The modeling window begins in 2003 because the notebook EDA found qualifying coverage to be consistent from that season. Earlier data remains part of the audit and descriptive context but is not included in this qualifying-based model.

Data preparation and model design

The raw relational dataset contains 14 tables. Cleaning replaces the literal missing-value sentinel \N, applies semantic types to dates and durations, and preserves the original tables. Repeated (raceId, driverId) result rows are reduced to one row by retaining the best (lowest) positionOrder, matching the podium target definition. This removed 91 extra rows from results.

Model features

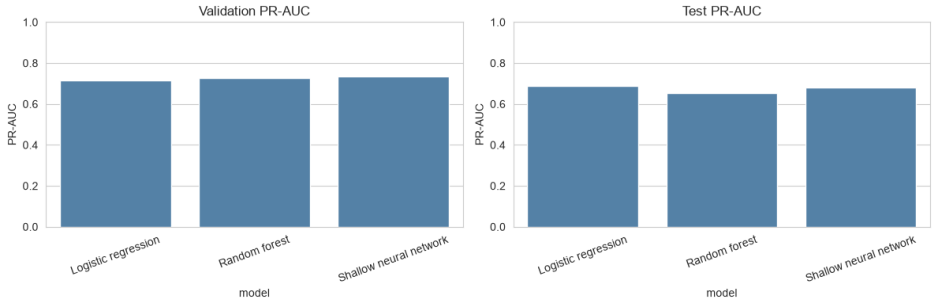
Feature group	Included features	Why available before race start
Grid and qualifying	Grid, qualifying position, Q1/Q2/Q3 times, session-missing flags	Grid and qualifying sessions are complete before the race starts.
Prior driver form	Driver podium rate in previous five starts	Uses earlier race results only; shifted to exclude the current race.
Prior constructor form	Constructor podium rate over previous five race entries	Aggregated by team and race, then shifted to earlier races.
Prior circuit record	Driver podium rate at this circuit in earlier races	Historical circuit experience only; current result excluded.
Context and identity	Season, circuit ID, driver ID, constructor ID	Known at prediction time; unseen categories are ignored by the encoder.

Chronological evaluation

Training uses 2003-2018 (6,411 rows), validation uses 2019-2021 (1,200 rows), and the unseen test uses 2022-2024 (1,359 rows). Podium prevalence is approximately 14-15% in all three splits. Splitting by season keeps each race wholly within one split. Imputation, scaling, and categorical encoding are fit on training data only.

The model comparison includes logistic regression, a random forest, and a shallow feed-forward neural network with one hidden layer. ROC-AUC, PR-AUC, and F1 are reported; the F1 threshold is selected on validation only. PR-AUC is the primary ranking metric because podiums are the minority class.

Validation and test PR-AUC comparison

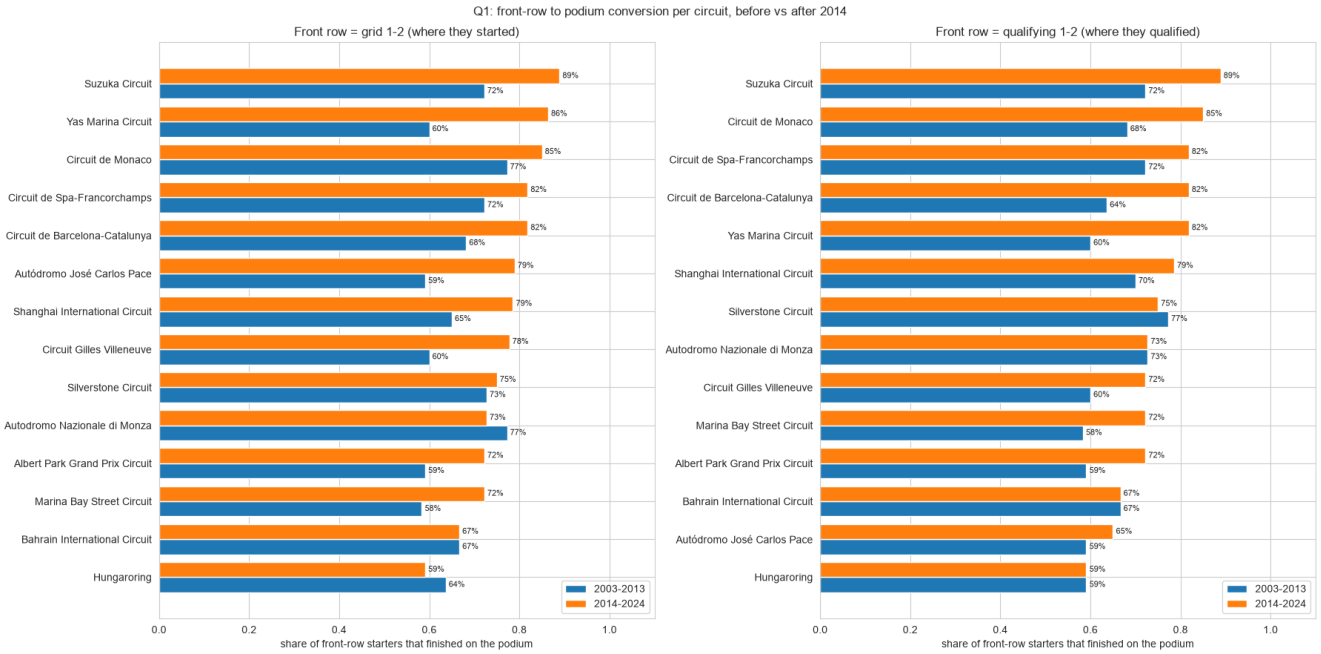


The chart above compares validation and test PR-AUC. Full train/validation/test metrics are shown in the model table on page 6.

Data-engineering question 1: front-row conversion

Question. Which circuits convert a front-row start into a podium most often, and did the ranking change after 2014? The analysis distinguishes qualifying position (P1-P2 in qualifying) from actual grid position (P1-P2 at race start).

Answer. Using actual grid position, front-row-to-podium conversion increased from 68% in 2003-2013 to 76% in 2014-2024. Since 2014, Suzuka (89%), Yas Marina (86%), and Monaco (85%) ranked highest among circuits with enough starts. Before 2014, Monaco and Monza led at 77%, followed by Silverstone, Suzuka, and Spa at about 72-73%. The ranking changed; Monaco remained near the top in both periods.



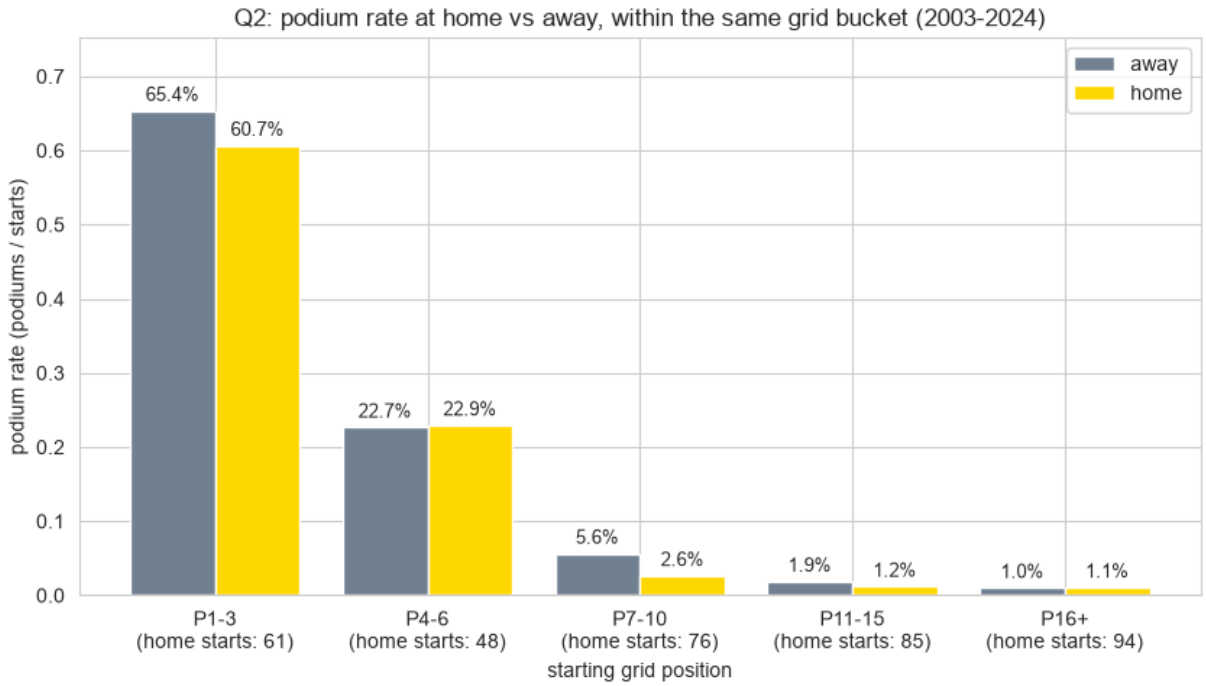
The comparison includes circuits raced in both periods with at least 10 front-row starts in each. Qualifying and grid definitions disagreed in about 4% of the front-row cases: grid penalties moved some front-row qualifiers back, while other drivers moved forward. The qualifying-based rate therefore includes penalty effects.

Grain and cutoff: one row is one driver in one race. The analysis uses 2003-2024, where qualifying records are consistently available. Small circuit samples make small percentage differences uncertain.

Data-engineering question 2: home-country starts

Question. Do drivers podium more often at a circuit in their home country after comparing drivers with similar starting grid positions?

Answer. The analysis finds no evidence of a home advantage after stratifying by grid buckets (P1-3, P4-6, P7-10, P11-15, P16+, and pit lane). Home and away rates were close within each bucket. Actual home starts yielded 52 podiums, compared with about 57.5 expected if home starters matched the away rates in the same grid buckets. The difference is small and is not evidence that home drivers perform worse.



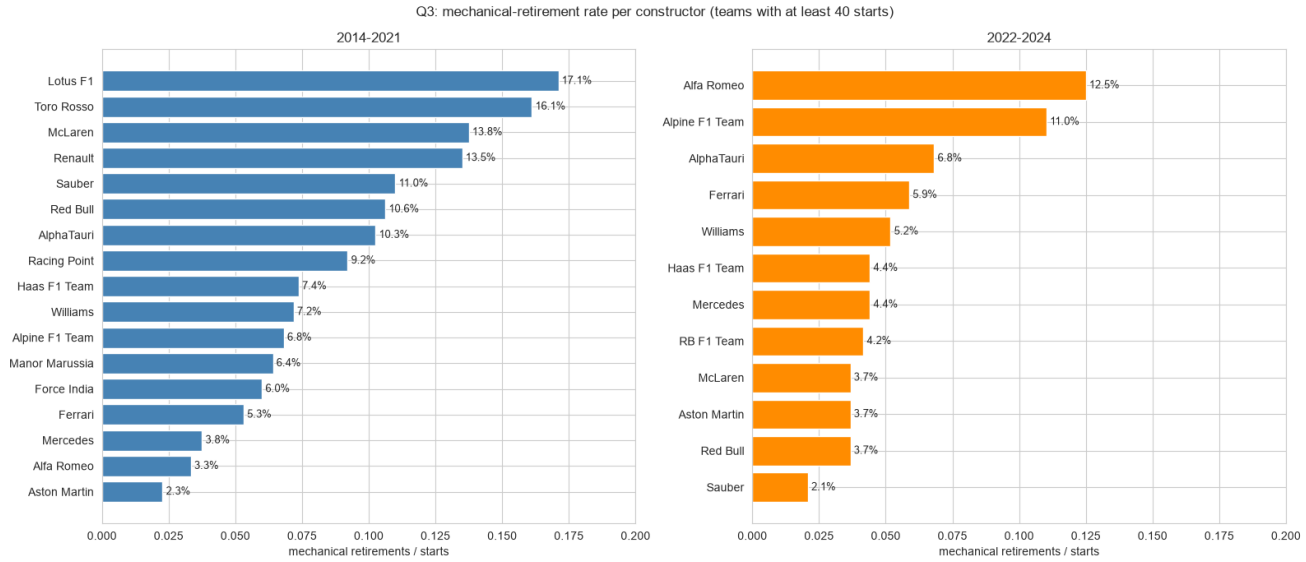
Without grid control, home and away podium rates were also similar (14.2% versus 14.3%). Home status is approximated from driver nationality mapped to country, so it is a proxy rather than verified residence or birthplace. Only 365 of 8,970 starts were classified as home starts; some nationalities could not be mapped to a country with a circuit in this period.

Grain and cutoff: one row is one driver in one race, 2003-2024. Grid position is known before the race; podium is the post-race outcome. The comparison is descriptive and does not adjust for all driver, team, or era differences.

Data-engineering question 3: mechanical retirements

Question. Which constructors had the highest mechanical-retirement rates in 2014-2021 versus 2022-2024?

Answer. In 2014-2021, Lotus F1 had the highest listed mechanical-retirement rate (17%), followed by Toro Rosso (16%), McLaren (14%), and Renault (14%); Mercedes was lowest at 3.8%. In 2022-2024, Alfa Romeo was highest (12.5%), followed by Alpine (11%); Red Bull, McLaren, and Aston Martin shared the lowest rate at 3.7%. Across the eras, the overall rate fell from 9.2% to 5.7%.



The numerator is starts whose status matches a 37-item list of component failure statuses; the denominator is starts. Crashes, driver/rule reasons, unclear retirements, and ambiguous tyre/wheel-related statuses are excluded. Teams with fewer than 40 starts in an era are filtered out. Constructor names are kept as recorded, so rebrands appear separately. The status is known after the race, so this is a descriptive analysis and is never a model feature.

Model comparison and experiments

The table reports all three metrics for training, validation, and test. Each model has its own F1 threshold selected on validation: logistic regression 0.340, random forest 0.605, and neural network 0.345.

Model	Split	ROC-AUC	PR-AUC	F1	Validation threshold
Logistic regression	Train	0.936	0.701	0.674	0.340
Logistic regression	Validation	0.930	0.715	0.717	0.340
Logistic regression	Test	0.931	0.690	0.562	0.340
Random forest	Train	0.976	0.873	0.762	0.605
Random forest	Validation	0.933	0.729	0.733	0.605
Random forest	Test	0.930	0.653	0.674	0.605
Shallow neural network	Train	0.941	0.719	0.680	0.345
Shallow neural network	Validation	0.923	0.734	0.725	0.345
Shallow neural network	Test	0.926	0.679	0.605	0.345

The neural network narrowly led validation PR-AUC (0.734). Test performance is lower and the ranking changes: logistic regression had the strongest test PR-AUC (0.690), while the random forest had the highest test F1 at its validation threshold (0.674). We do not use these test rankings to retune or switch models. The random forest training PR-AUC of 0.873 versus validation 0.729 suggests overfit.

Feature-group ablation

Removed group	Validation PR-AUC	Test PR-AUC	Test F1
None (full logistic baseline)	0.715	0.690	0.562
Grid + qualifying	0.627	0.518	0.471
Driver prior form	0.706	0.691	0.549
Constructor prior form	0.722	0.685	0.667
Driver prior circuit record	0.717	0.691	0.548
Season and entity IDs	0.721	0.700	0.670

Removing grid and qualifying lowered validation PR-AUC by 0.088, the clearest contribution. Removing driver prior form lowered it by about 0.010. Removing the other groups slightly raised validation PR-AUC; those small differences may be noise. The test columns are final reporting only, not the basis for feature selection.

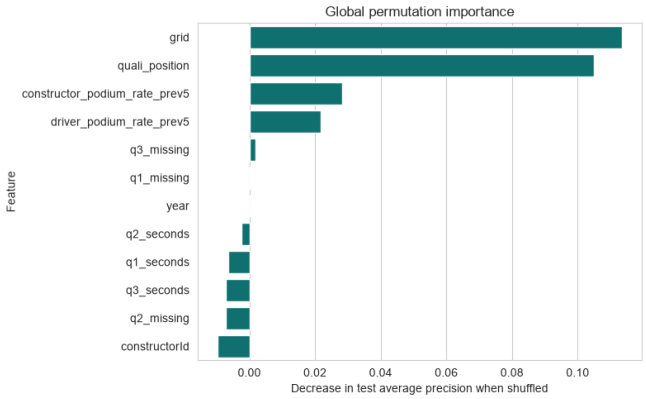
Preprocessing-order experiment

Median imputation followed by scaling reached validation PR-AUC 0.7153 and F1 0.7173. Scaling observed values then filling missing standardized values with zero reached PR-AUC 0.7158 and F1 0.7154. The 0.0005 PR-AUC difference is negligible; median-impute-then-scale is retained as the main sequence. Test PR-AUCs were 0.6897 and 0.6905 respectively.

Explainability, inference, and limitations

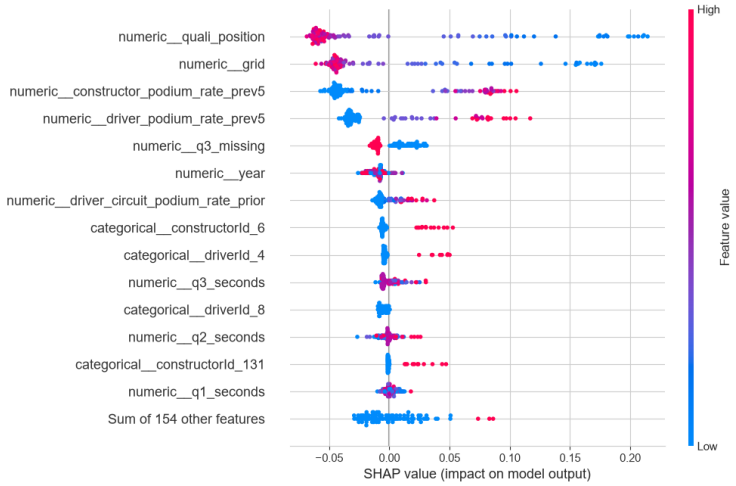
Explainability uses the random forest, the higher-PR-AUC statistical model on validation. Global permutation importance and SHAP are model-reliance explanations, not causal estimates.

Global permutation importance



Shuffling grid reduced average precision by 0.113 on the sampled test rows; shuffling qualifying position reduced it by 0.105. Constructor and driver prior form followed at 0.028 and 0.022. Near-zero or slightly negative values for other correlated features indicate little added permutation value in this model/sample, not a harmful causal effect.

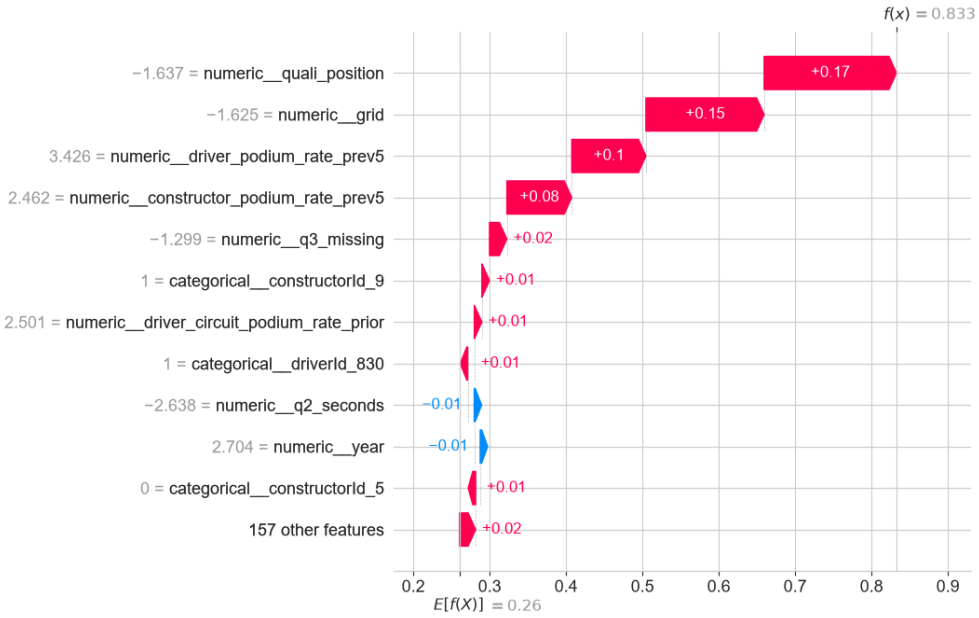
SHAP global summary



The strongest mean absolute SHAP contributions were qualifying position, grid, constructor prior podium rate, and driver prior podium rate. Better (lower) qualifying/grid positions and stronger prior form generally push this model output toward a podium.

Local example and inference

The illustrated 2023 test row qualified and started P1, had strong prior driver/team form, and finished on the podium. The random forest assigned a podium probability of 0.833. In the local SHAP waterfall, qualifying position (+0.17) and grid (+0.15) were the largest positive contributions, followed by driver prior form (+0.10) and constructor prior form (+0.08).



The inference function accepted a complete test record and returned probability 0.706 (predicted podium at the validation threshold 0.605). Removing qualifying position, Q2/Q3 times, and driver prior form from that same record lowered the probability to 0.472 and changed the thresholded prediction to no podium. Missing values can therefore materially affect an individual prediction.

Limitations

- The model covers 2003-2024. Regulation changes, team rebrands, and driver turnover can cause temporal drift.
- Podiums are a minority class; F1 depends on the threshold, which was selected on validation.
- Unseen drivers and constructors have no learned ID-specific history. Their rolling form starts missing and is imputed by the training pipeline.
- Home country is approximated using nationality and a hand-maintained mapping.
- Mechanical retirement depends on a manually specified status list; ambiguous tyre/wheel events are excluded.
- Permutation importance and SHAP explain model behavior, not causes of race outcomes.

Deliverable companion: the run-all Jupyter notebook contains data audit, cleaning, EDA, data-engineering answers, feature construction, model training, experiments, XAI plots, and the inference function.